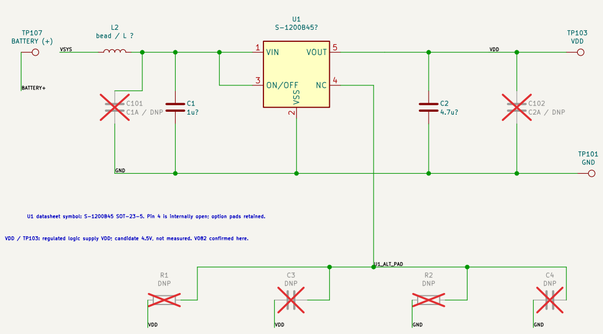
01 POWER / INPUT FILTER

H01: likely S-1200B45 LDO. Nominal 4.5 V is a candidate, not a measured rail.



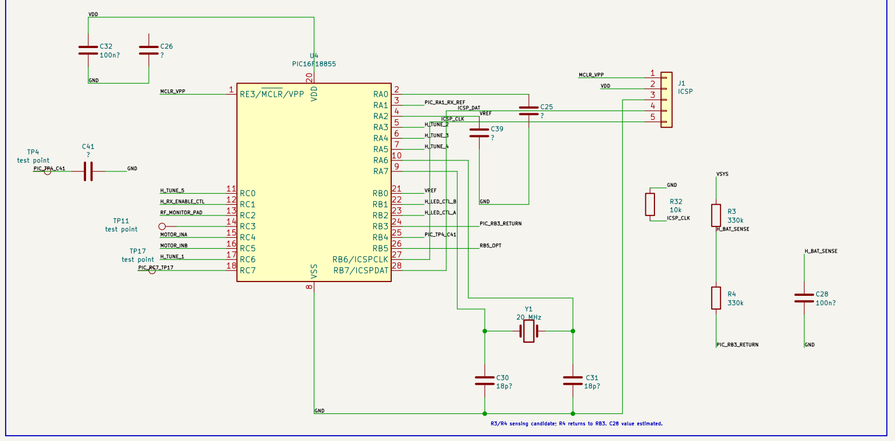
U1 datasheet symbol: S-1200B45 SOT-23-5. Pin 4 is internally open; option pads retained.

VDD / TP103: regulated logic supply VDD: candidate 4.5V, not measured. VDB2 confirmed here.

Empty options: R1/C3 free ends join regulated VDD: R2/C4 free ends join GND.

02 PIC CONTROLLER / CLOCK / ICSP

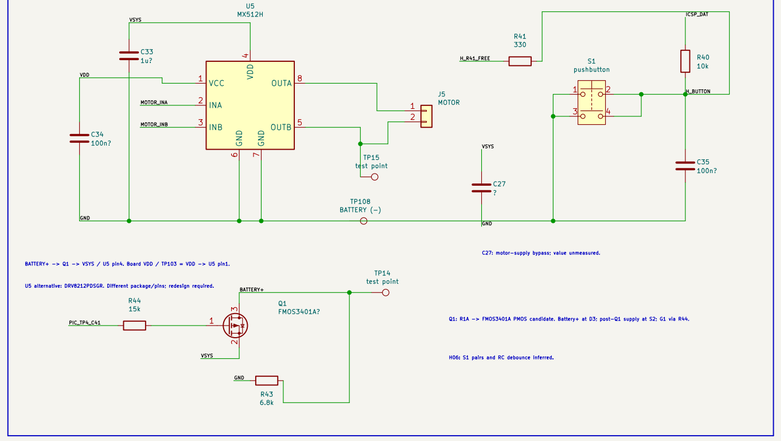
User: RA3/R14, RA4/R15, RA5/R16, RC0/R11, RC6/R13, RC2 -> TP3; RB0 -> VREF, RC1 -> R18 via VD63-VD70.



R3/R4 sensing candidate: R4 returns to RB3. C28 value estimated.

03 MOTOR / BUTTON / BATTERY INTERFACE

User: PIC15/RC4 -> INA pin2: PIC16/RC5 -> INB pin3. Motor output polarity remains inferred.



BATTERY+ -> Q1 -> VSYS / U5 pin4, Board VDD / TP103 = VDD -> U5 pin1.

U5 alternative: DRV8212PDSGR. Different package/pins; redesign required.

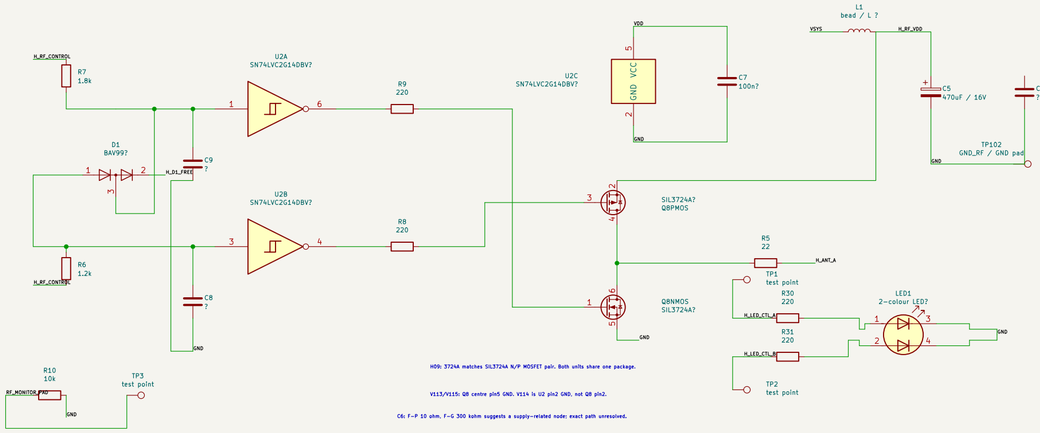
C27: motor-supply bypass: value unmeasured.

Q1: R1A -> FMOS3401A PMOS candidate. Battery+ at D3: post-Q1 supply at S2: G1 via R44.

H06: S1 pairs and RC debounce inferred.

04 ANTENNA EXCITATION / LOGIC / INDICATOR

H07: C14R inverter candidate, Q8 source rail inferred: C6 destination unresolved.



H09: 3724A matches SIL3724A N/P MOSFET pair. Both units share one package.

V113/V115: Q8 centre pin5 GND. V114 is U2 pin2 GND, not Q8 pin2.

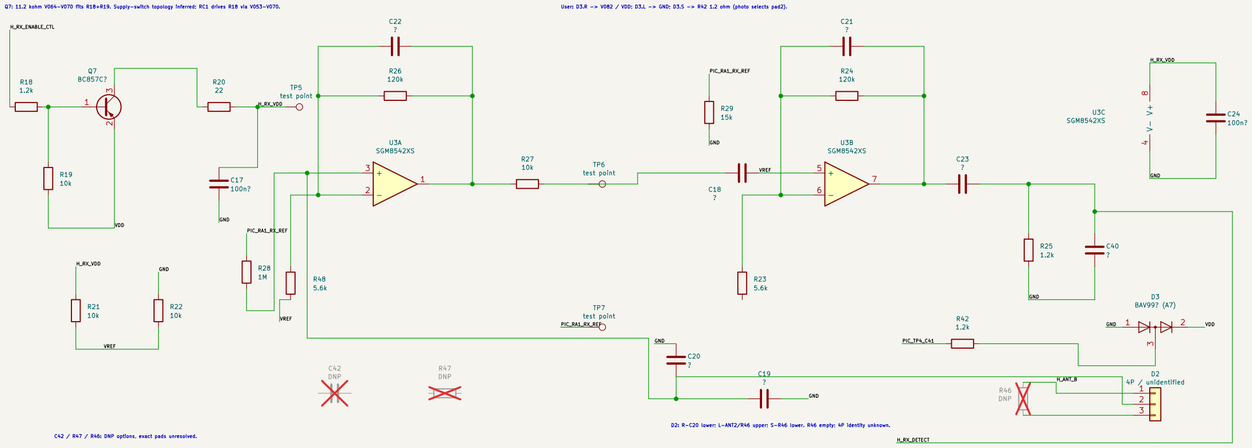
C6: F-P 10 ohm, F-G 300 kohm suggests a supply-related node; exact path unresolved.

05 RF RECEIVE / GAIN / ENVELOPE

H10–H12: working topology around photographed 120k feedback resistors. Filters, bias, and interstage routes inferred.

Q7: 11.2 kohm VD64–VD70 ffs R18+R19. Supply-switch topology Inferred: RC1 drives R18 via VD63–VD70.

User: D3.R -> VD82 / VDD: D3.L -> GND: D3.C -> R42 1.2 ohm (photo selects pad2).



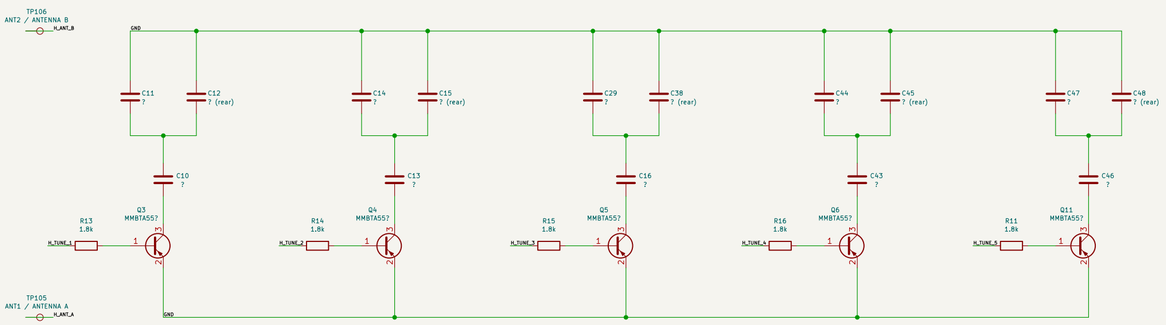
C42 / R47 / R46: DNP options, exact pads unresolved.

U3 alternative: MCP6002-I/SN: same pin roles, electrical suitability to verify.

User: U3 pin5 / VD79 / VD80 = VREF (1 ohm): TP6 remains separate across C18.

06 SWITCHED ANTENNA CAPACITANCE

H13: 2H devices are probably PNP BJTs (MMBTA55 family; PBSS5140T is an alternative). Capacitor values unknown.



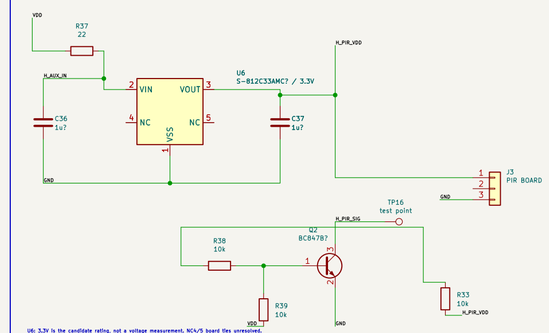
v0.9: user-reviewed GND returns replace previous antenna rails on this capacitor bank.

Local three-capacitor junctions retained. Hidden links from these midpoints to antenna remain unknown.

Do not infer complete tuning topology from local connections. Transistor identity/orientation remains provisional.

07 PIR / AUXILIARY RAIL

H14–H15: C2N supports S-812C33AMC 3.3V LDO. C/R3 ground supported by resistance.

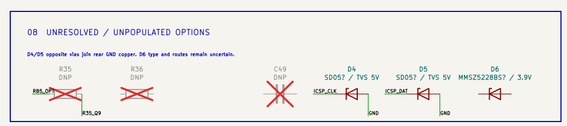


U6: 3.3V is the candidate rating, not a voltage measurement. NC4/5 board ties unresolved.

User: VD63–VD64 confirms R37/R39 feed an regulated VDD.

08 UNRESOLVED / UNPOPULATED OPTIONS

D4/D5 opposite vias join rear GND copper. D6 type and routes remain uncertain.



READING THIS WORKING RECONSTRUCTION

One sheet. Functional wiring is proposed: this is not a verified production circuit.

150 original main-board catalog entries, including DNP footprints and named pads.

H_* = inferred inter-block net. Other net names do not imply measurement.

? on a part value or pin function means an estimate or candidate identity.

Library pin numbers are mapped to photo pads in evidence/pin_crosswalk.json.

U2 and U3 use multiple symbol units on this same sheet, including supply units.

28 original local photo fragments retained: Q1.L–R43.2 withdrawn by user.

PIC GPIOs are intentionally unassigned where photographs cannot support a map.

DNP = unpopulated. An open pin means unresolved, not a certified no-connect.

PIR daughterboard internals are not yet traced: J3 represents its interface.

No capacitor value or rail voltage in this drawing was electrically measured.

See docs/RECONSTRUCTION.md for evidence, alternatives, and calculations.



Spare footprints / unassigned test pads

v0.9.13 | KiCad 10.0.5 | 2026-10-09

U3.5 = VREF; J3.3 = GND. Rejected R23/J3 guesses removed.

KiCad 10 libraries + four datasheet symbols | MCP authoring

Sheet: /

File: PetSafe_1001339.kicad_sch

Title: PetSafe 100–1339 R03 A | single-sheet reconstruction

Size: A2	Date: 2026-10-09	**Rev: 0.9.13**
KiCad E.D.A. 10.0.5		Id: 1/1